

AD 502 – Machine Learning

Pre-Reading Material – Lecture 28

Parameter Estimations: MLE, MAP

Introduction

In machine learning and statistics, a model often contains **unknown parameters** that need to be estimated from available data.

Parameter estimation is the process of finding suitable values for these unknown parameters so that the model represents the observed data effectively.

For example, in a Gaussian distribution, parameters such as the **mean** and **variance** need to be estimated from the available observations.

Two important approaches to parameter estimation are:

- **Maximum Likelihood Estimation (MLE)**
- **Maximum A Posteriori (MAP) Estimation**

Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation (MLE) is a method for estimating model parameters by finding the parameter values that make the observed data **most likely**.

In simple words:

MLE chooses the parameter values that best explain the observed data.

For example, suppose we have observations of students' marks and want to estimate the average performance of the class. MLE uses the observed data to determine the parameter value that makes those observations most likely under the assumed model.

Key idea:

Observed Data → Find parameters that best explain the data → Estimated Parameters

MLE mainly relies on the **observed data**.

Maximum A Posteriori (MAP) Estimation

Maximum A Posteriori (MAP) estimation extends the idea of MLE by considering both:

- The **observed data**
- Prior knowledge or assumptions about the parameters

In simple words:

MAP chooses parameter values that are supported by both the data and prior beliefs.

For example, if previous studies suggest that a parameter is likely to lie within a particular range, MAP can incorporate this information along with the current dataset.

Key idea:

Prior Knowledge + Observed Data → Estimated Parameters

MLE vs MAP

MLE	MAP
Uses observed data	Uses observed data + prior knowledge
Focuses on likelihood	Considers likelihood and prior belief
Does not explicitly use a prior distribution	Uses a prior distribution
Data-driven estimation	Data + prior-based estimation

MLE

MAP

Useful when sufficient data is available Useful when prior knowledge is available

Relationship with EM and GMM

Parameter estimation is important in **Gaussian Mixture Models (GMMs)**.

GMM needs to estimate parameters such as:

- Mean of each Gaussian component
- Spread of each component
- Relative contribution of each component

The **Expectation-Maximization (EM) algorithm** is commonly used to estimate these parameters in GMM.

Therefore, the concepts can be connected as:

GMM → Parameter Estimation → EM Algorithm → Estimated Model Parameters

MLE can be used as the underlying estimation principle in standard EM-based GMM training, while MAP estimation can incorporate prior information into the parameter estimation process.

Key Takeaways

- **Parameter estimation** finds unknown model parameters from data.
- **MLE** finds parameters that make the observed data most likely.
- **MAP** combines observed data with prior knowledge.
- **MLE is data-driven**, while **MAP uses data and prior information**.
- Parameter estimation is an important part of **GMM and EM**.

